

# Dinh Truong Lam

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## EXPERIENCE

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- **MoMo (M-Service)** Vietnam  
*Software Engineer II (Payment Platform)* May 2025 - Present
  - Engineered high-throughput checkout services handling core payment flows using **Java** and **Vert.x** (Event-driven, non-blocking I/O) to support **millions** of daily active users.
  - Slashed Payment ID generation latency by **60%** (from ~300ms down to ~120ms) by replacing per-item RabbitMQ async processing with a unified, low-latency gRPC synchronous call.
  - Ensured **99.99%** availability for critical payment flows by implementing Circuit Breakers to gracefully handle downstream service degradation (e.g., Gift, Source of Funds).
  - Resolved concurrency race conditions in the complex Coin Redemption flow by implementing **Redis Distributed Locks**, preventing double-spending and ensuring data consistency.
  - Enhanced system observability and monitoring by utilizing **Kafka** and **Splunk** for distributed tracing, and actively reducing **Grafana** metric cardinality to optimize dashboard performance.
  - Owned the end-to-end delivery of the "Source-based Gift Recommendation" feature, utilizing Unleash for feature-flagging and safe canary rollouts while collaborating seamlessly with cross-functional teams.

**Technologies used:** Java, Vert.x, gRPC, RabbitMQ, Redis, Kafka, Splunk, Unleash.

- **Reksoft International** St. Petersburg, Russia  
*Software Engineer (Outsource)* Jun 2024 - Apr 2025
  - Engineered core microservices for a large-scale B2B Order Management System (OMS) using **Java** and **Spring Boot**, successfully processing over **20,000** daily wholesale orders for enterprise clients.
  - Eradicated inventory overselling during peak purchasing hours by implementing strict Database Locking mechanisms (Pessimistic/Optimistic Locks) in **PostgreSQL**, ensuring **100%** transactional consistency.
  - Integrated 3rd-party logistics (3PL) APIs (e.g., DHL, FedEx) for shipment generation, applying **Resilience4j** Circuit Breakers to gracefully handle network instability and prevent cascading failures during vendor downtimes.
  - Architected a secure Webhook ingestion pipeline for real-time delivery tracking, implementing HMAC signature validation for security and utilizing **Apache Kafka** to decouple payload processing, ensuring **<10ms** acknowledgment to external providers.

**Technologies used:** Java, Spring Boot, PostgreSQL, Kafka, Resilience4j, REST API, Webhooks.

- **ITMO University Lab** St. Petersburg, Russia  
*Software Engineer* Oct 2022 - Mar 2024

**Project: Smart City Environmental Data Platform**

  - Architected a scalable environmental monitoring backend (Pilot phase) serving **50+** IoT stations, reliably processing and storing **~15 million** time-series records daily.
  - Designed an Event-Driven Ingestion Pipeline using **Apache Kafka** to absorb sudden traffic bursts, completely eliminating database bottlenecks during peak telemetry syncs.
  - Implemented the Claim Check Pattern with **MinIO** to process nightly **1GB+** CSV batch files, successfully preventing Out-Of-Memory (OOM) crashes and keeping server RAM footprint under **300MB** during ETL processes.
  - Applied Polyglot Persistence architecture, routing raw time-series data to **Cassandra** for fast, append-only writes, while utilizing **PostgreSQL** for structured, aggregated daily reporting.
  - Engineered fault-tolerant background workers using **Redis Distributed Locks (SETNX)** and Database Idempotency keys, preventing race conditions and duplicate processing across multiple instances.
  - Slashed dashboard API latency from ~5 seconds to **<50ms** by implementing a Cache-Aside pattern in **Redis** for frequently accessed analytical queries.

**Technologies used:** Java, Apache Kafka, MinIO, Cassandra, PostgreSQL, Redis, Docker.

## SKILLS

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- **Programming Languages:** Java, Golang, C++, TypeScript, JavaScript, Python.

- **Frameworks & Technologies:** Spring Boot, Vert.x, gRPC, REST API, Hibernate/JPA, Resilience4j.
- **Databases & Infrastructure:** PostgreSQL, Cassandra, Redis, Apache Kafka, RabbitMQ, MinIO, Docker, Splunk, Grafana.
- **AI-Assisted Development:** Highly proficient in utilizing AI coding assistants (**Cursor**, **GitHub Copilot**, **Claude (Opus/Sonnet)**, **Codex**) to accelerate development cycles, refactor complex codebases, and optimize algorithms.
- **Spoken Languages:** English (**C1 Advanced** - Proficient in all 4 skills, CEFR standard), Vietnamese (Native).

EDUCATION

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- **ITMO University** Saint Petersburg, Russia  
*Bachelor of Science in Software Engineering; GPA: 3.2/4.0* Sep 2019 – May 2025
- **University of Engineering and Technology, Vietnam National University** Hanoi, Vietnam  
*Bachelor of Computer Science* Aug 2019 – Sep 2019
- **Thai Nguyen High School for the Gifted** Thai Nguyen, Vietnam  
*Informatics class* Sep 2016 – Jun 2019

AWARDS & ACHIEVEMENTS

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- **First Prize in Russia Information Technology Olympic** 2019  
*Organized by the Vietnamese and Russian ministries of education*
- **Bronze Medal, Vietnam National Olympiad in Informatics (Grade 12)** 2019
- **Third Prize, CodeTour (National Programming Contest)** 2019
- **2x Gold Medals, Hung Vuong Summer Camp in Informatics** 2018

ACTIVITIES

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- **ITMO Brain Club** Nov 2021 - Jun 2022  
*Lecturer/Teaching Assistant: Taught programming foundations and data structures to Vietnamese students studying at ITMO.*